

Topics of today's class

Last class:

- Chapter 2
 - Nominal and Real GDP and Price Indices
 - Savings, wealth and capital
 - Labor market measurement

Today's class:

- Chapter 4
 - Behavior of the representative consumer

Chapter 4

Consumer and Firm Behavior: The Work-Leisure Decision and Profit Maximization

John Keynes' Prediction in 1930

- In 1930, British economist John Keynes wrote an [essay](#) called "Economic Possibilities for Our Grandchildren." In the essay, he made the now-famous assertion that his grandchildren's generation (meaning people in the workforce today) would only work 15 hours a week.
- Gallup [reports](#) that the average full-time American put in nearly 47 hours a week in 2014.
- Why Keynes was so far off-base?

Representative Consumer

- Consumer's preferences.
- Consumer's budget constraint.
- Consumer's optimization problem.
- How does the consumer respond to:
 - (i) an increase in non-wage income;
 - (ii) an increase in the market real wage rate?

Preference – an example

- Ben is a tennis coach
 - He earns \$35 per hour by giving lesson.
 - But his real passion is skydiving.
- Ben's decision is: how many hours to work each year?
 - cost of working an extra hour: loss of an hour of skydiving.
 - benefit of working an extra hour: \$35 in income

Preference

- 2 goods:
 - consumption c
 - leisure l : economists use the term leisure for all off-the-job activities
- Utility Function

$$U(c, l)$$

Preference

- We say that (c_1, l_1) is strictly preferred by the consumer to (c_2, l_2) if

$$U(c_1, l_1) > U(c_2, l_2)$$

- and the consumer is indifferent between the two consumption bundles if

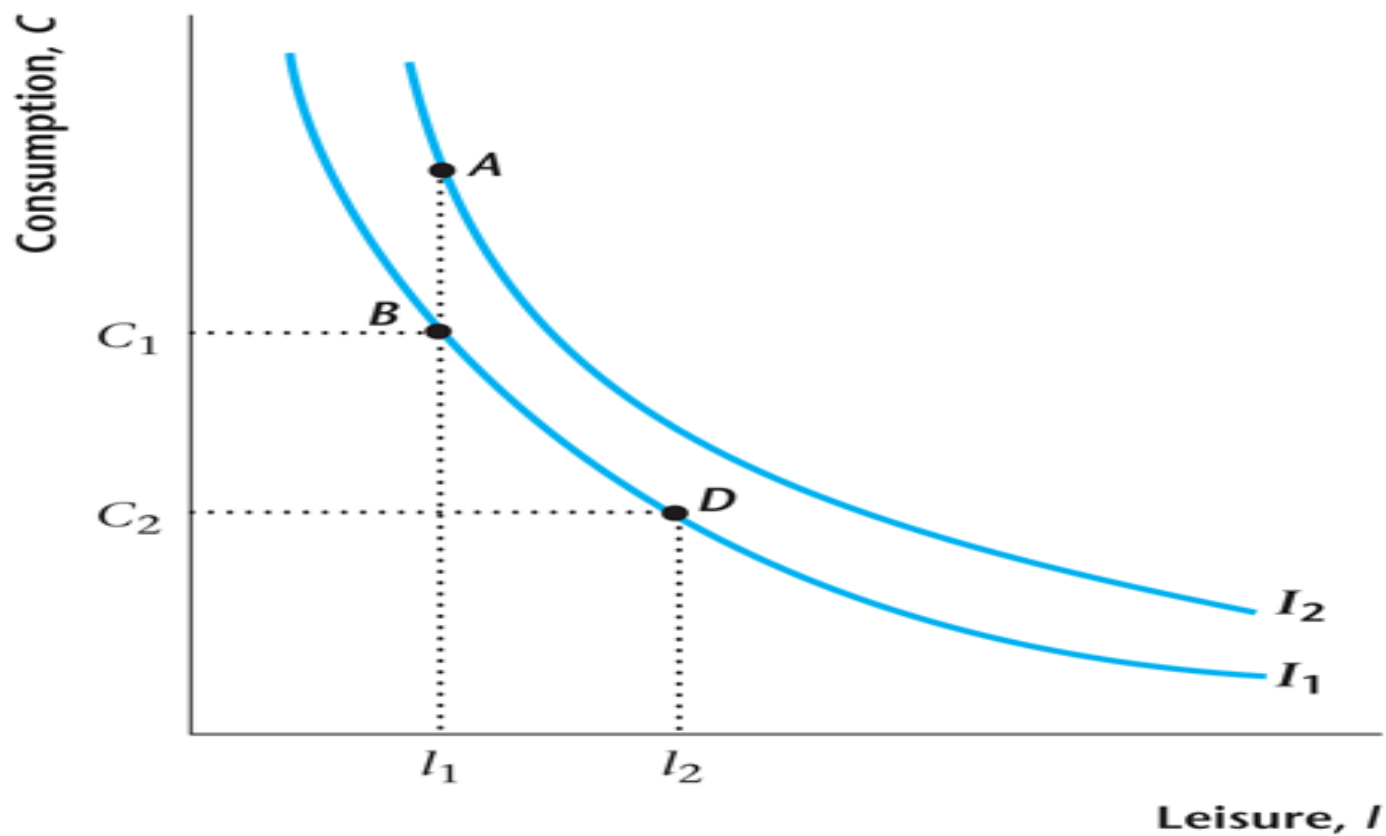
$$U(c_1, l_1) = U(c_2, l_2)$$

Indifference curves

- An indifference curve connects a set of points, with these points representing consumption bundles among which the consumer is indifferent.
 - *More is always preferred to less;*
 - *The consumer likes diversity in consumption bundle;*
 - *Consumption and leisure are normal goods.*

Figure 4.1

Indifference Curves

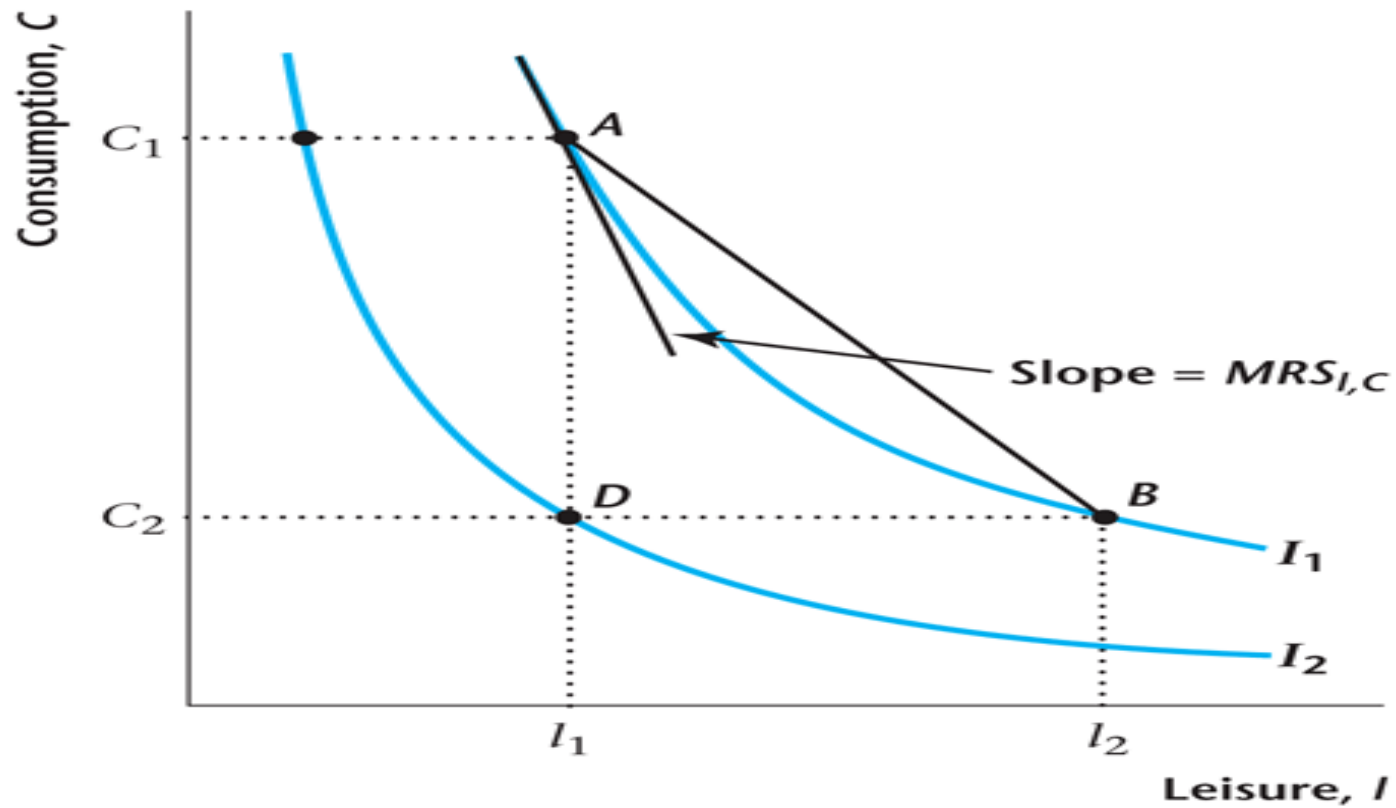


Indifference Curves

- An indifference curve slopes *downward* (more is preferred to less).
- An indifference curve is *convex* (the consumer prefers diversity in his or her consumption bundle).

Figure 4.2

Properties of Indifference Curves



Marginal rate of substitution

- The marginal rate of substitution of leisure for consumption is the rate at which the consumer is just willing to substitute leisure for consumption goods.
- $MRS_{l,c} = -[\text{the slope of indifference curve passing through the bundle } (c, l)]$

The Consumer's Time Constraint

l : leisure

N^s : labor supply or time spent working

h : hours of time available.

$$l + N^s = h,$$

The Consumer's Budget Constraint

- Consumption is equal to total wage income, plus dividend income, minus taxes.

$$C = wN^s + \pi - T,$$

The Consumer's Budget Constraint, Accounting for the Time Constraint

$$C = w(h - l) + \pi - T.$$

$$C + wl = wh + \pi - T.$$

$$C = -wl + wh + \pi - T,$$

Figure 4.3

Representative Consumer's Budget Constraint when $T > \pi$

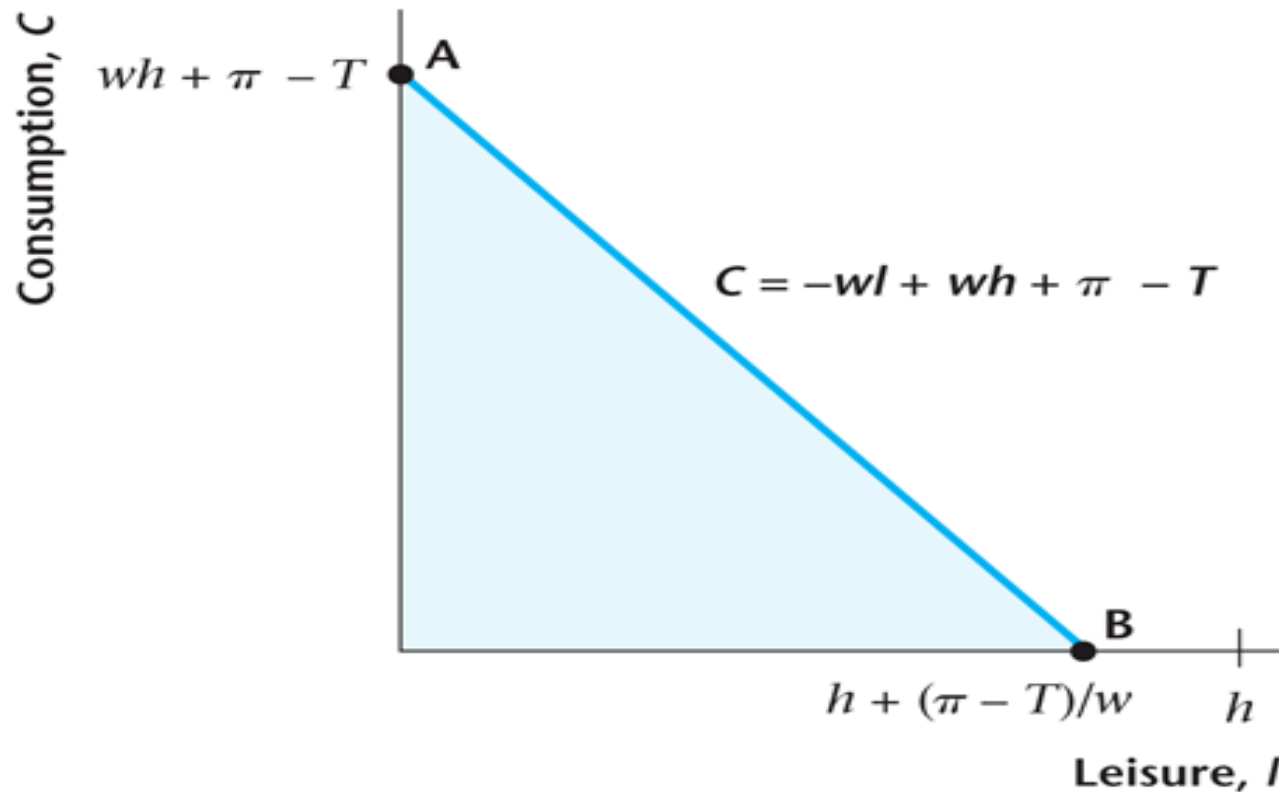
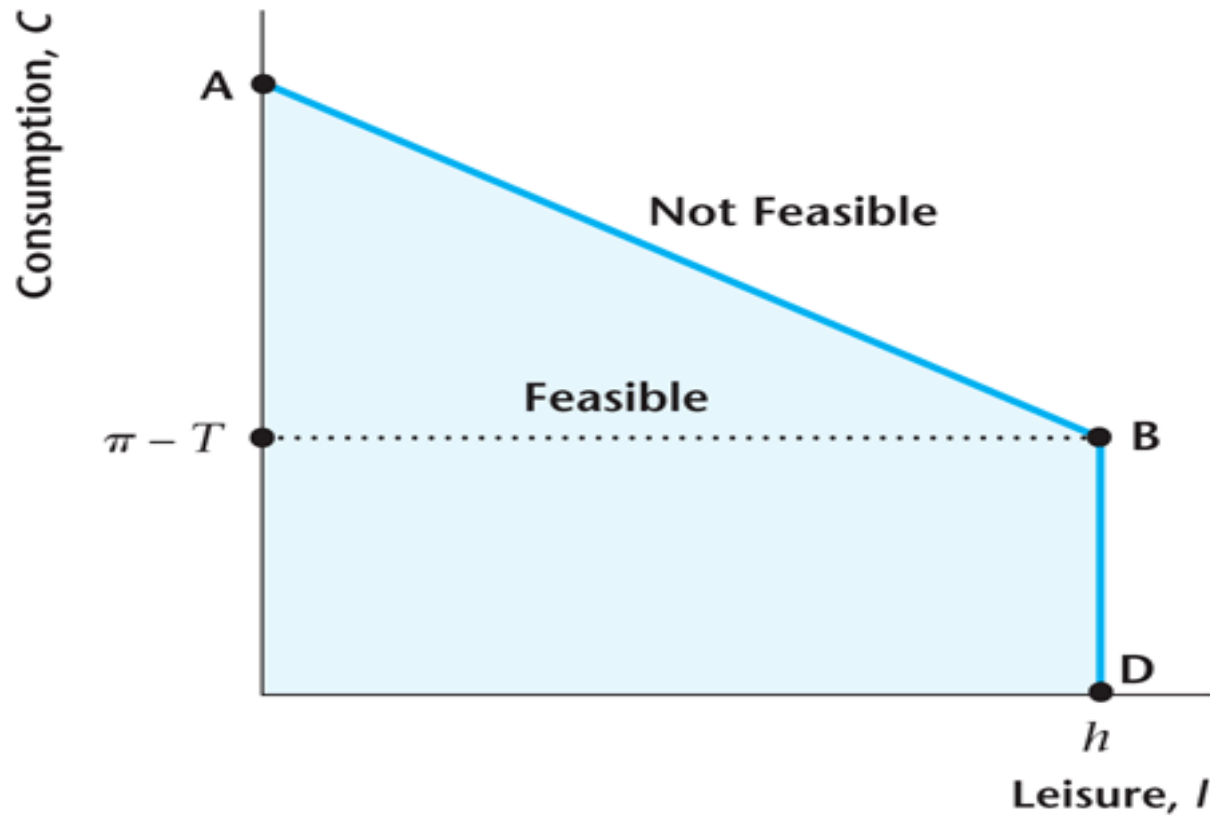


Figure 4.4

Representative Consumer's Budget Constraint when $T < \pi$

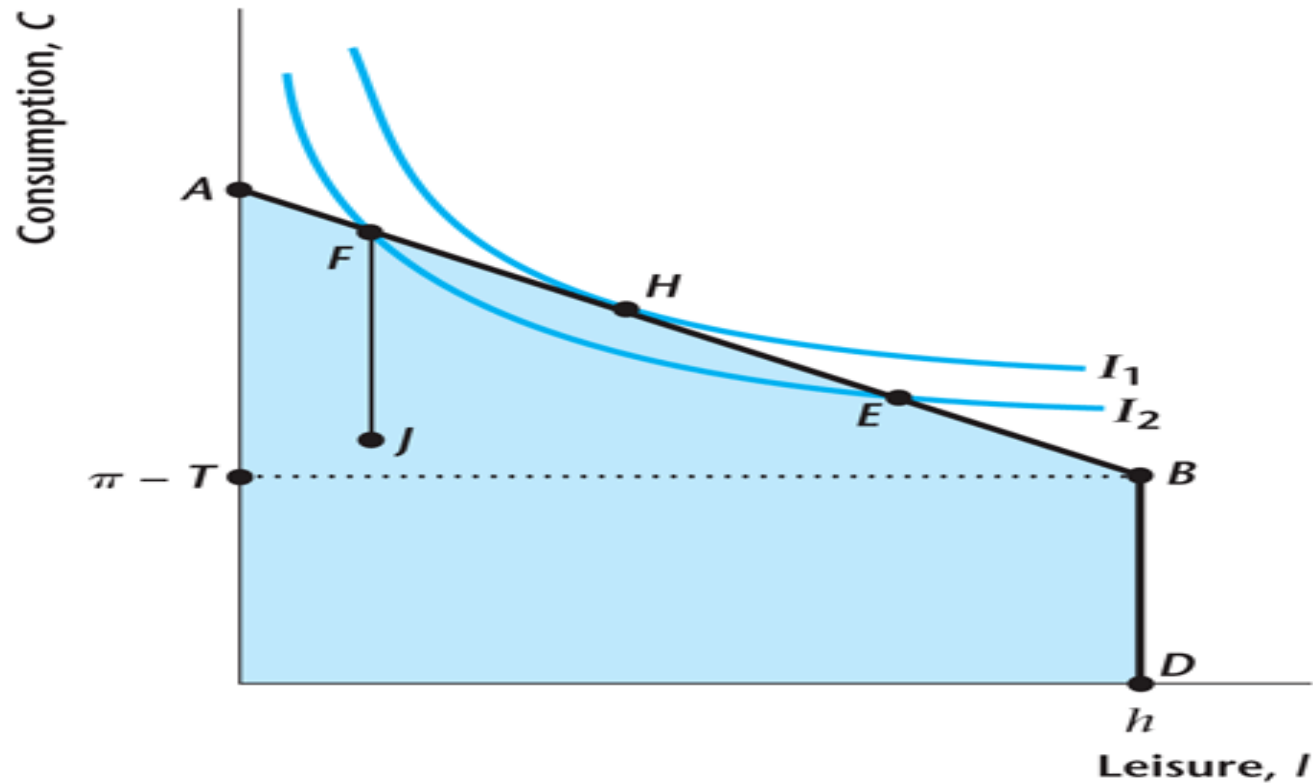


Consumer Optimization

The consumer chooses the consumption bundle that is on his or her highest indifference curve, while satisfying his / her budget constraint.

Figure 4.5

Consumer Optimization

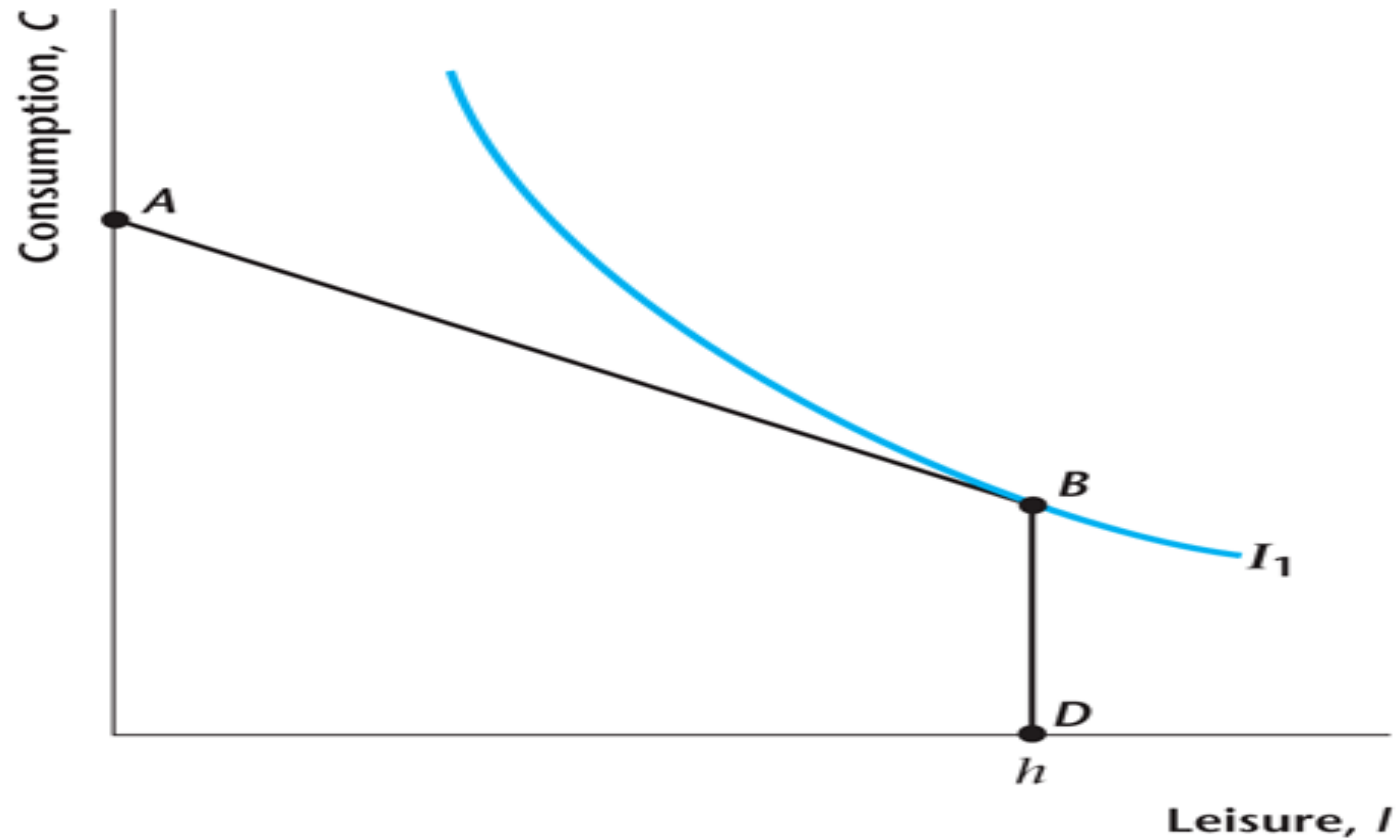


Optimization Implies:

The marginal rate of substitution of leisure for consumption equals the real wage.

Figure 4.6

The Representative Consumer Chooses Not to Work



Thought experiments

- Impact of a tax decrease.
- Impact of a wage increase.

Income and substitution effect

- Decomposition of the effect of policy into:
 - **Income Effect:** what is the impact on the budget constraint of the consumer?
 - **Substitution Effect:** what is the impact on the cost/benefit of working an additional hour?

Income and substitution effect - example

Pure Income Effect:

- Randy bought 1,000 shares of Apple at \$90.2 in early 2009.

“Pure” Substitution Effect:

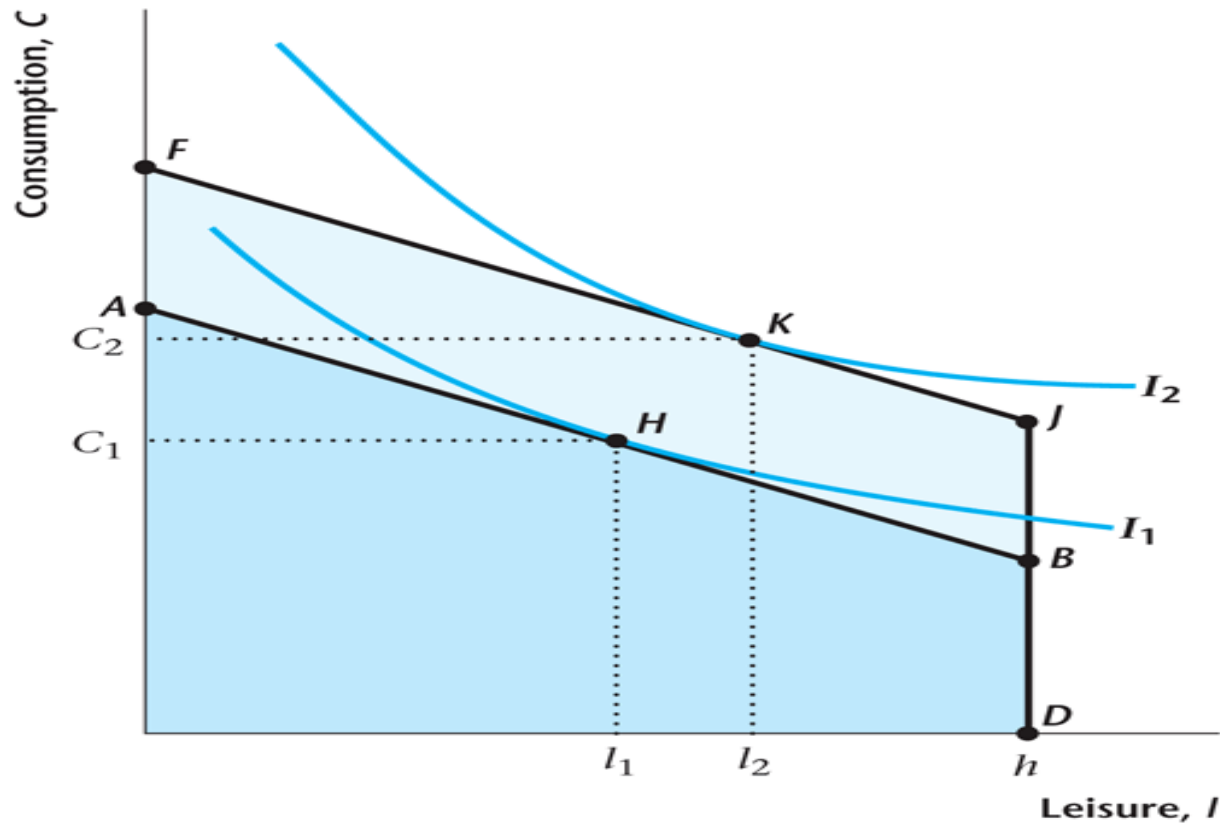
- a one-day rise in the real wage
 - Randy works for \$35 an hour. Mark wants to take a private lesson and is willing to pay \$350 per hour.

Real Dividends or Taxes Change for the Consumer

- Assume that consumption and leisure are both normal goods.
- An increase in dividends or a decrease in taxes will then cause the consumer to increase consumption and reduce the quantity of labor supplied (increase leisure).
- Pure income effect.

Figure 4.7

An Increase in $\pi - T$ for the Consumer



An Increase in the Market Real Wage Rate

- Ben works for \$35 at his current tennis club. Another tennis club offers him \$50 per hour. Will Ben increase his number of hours worked?
 1. the reward to working is greater: Ben is tempted work more.
 2. But: if Ben was working 10 hours a week before, he was earning \$350. Now he can earn the same income by working just 7 hours.

An Increase in the Market Real Wage Rate

- This has income and substitution effects.
- ***Substitution effect***: the price of leisure rises, so the consumer substitutes from leisure to consumption.
- ***Income effect***: the consumer is effectively wealthier and, since both goods are normal, consumption increases and leisure increases.
- Conclusion: Consumption must rise, but leisure may rise or fall.

Figure 4.8

Increase in the Real Wage Rate—Income and Substitution Effects

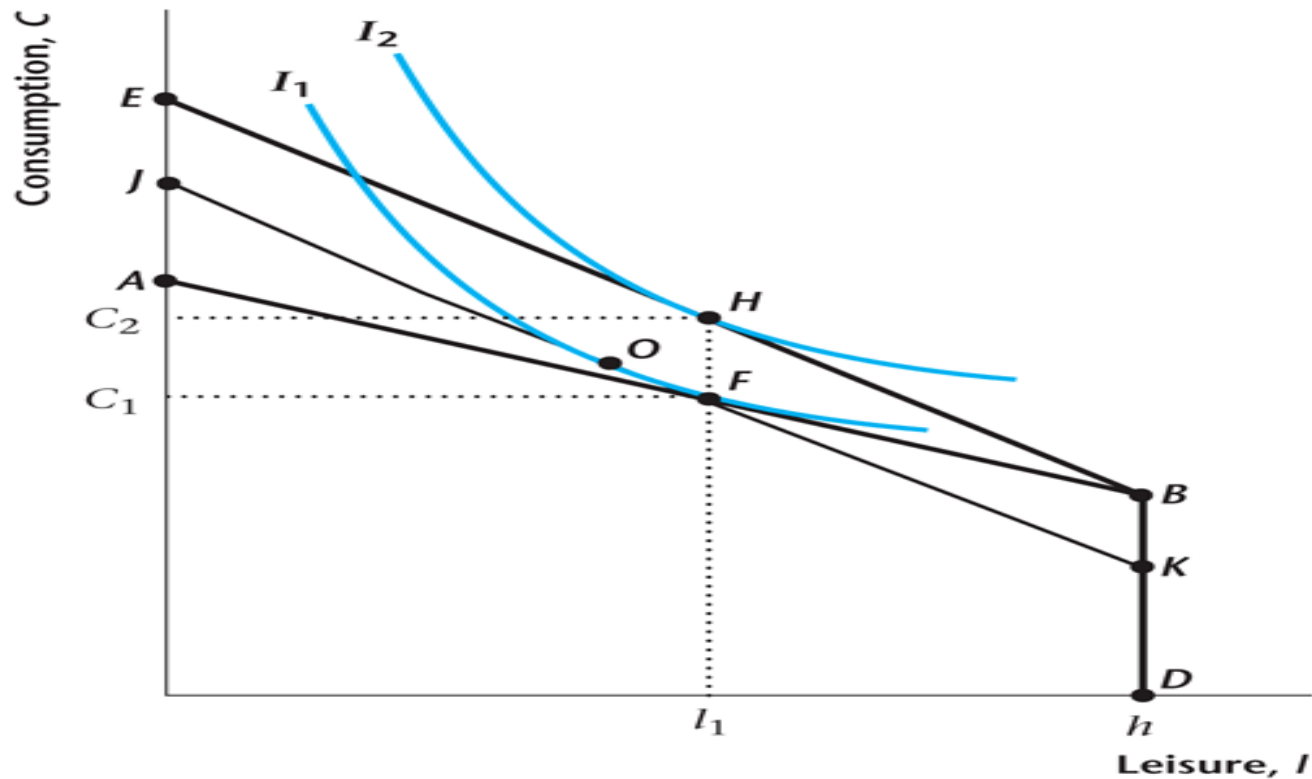


Figure 4.9

Labor Supply Curve

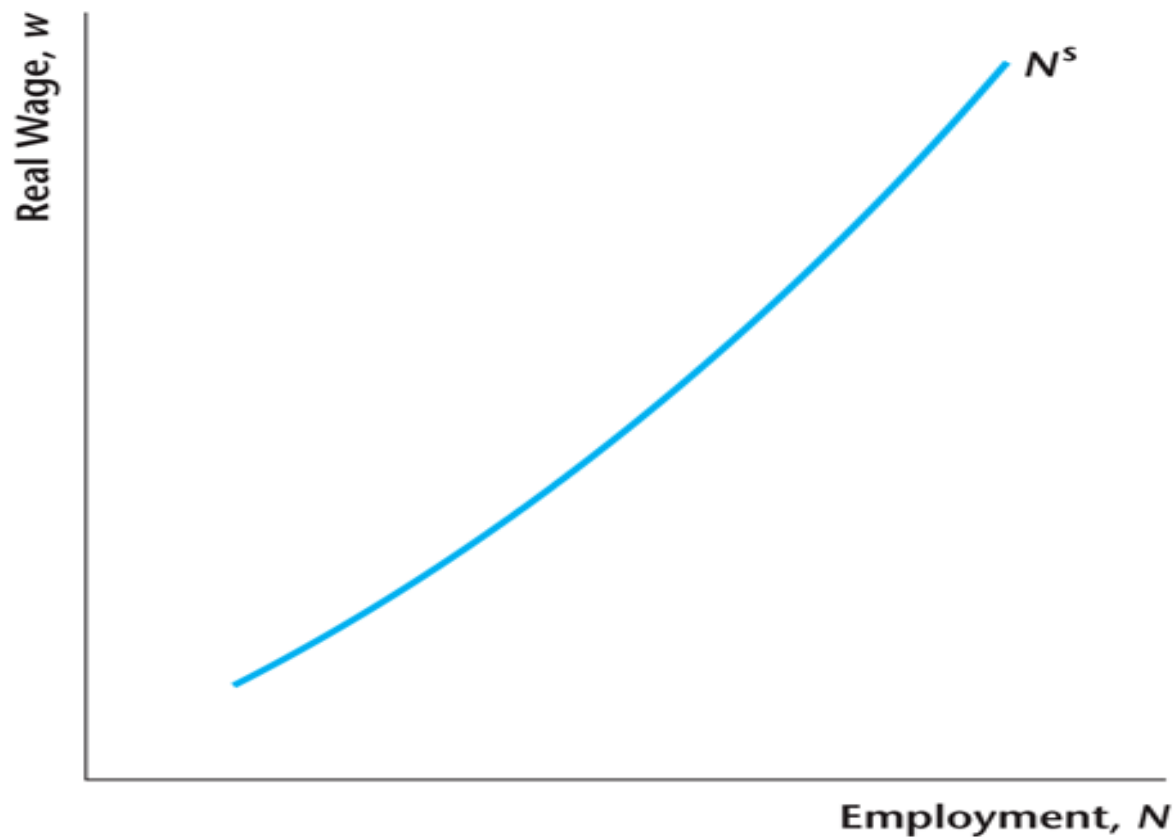


Figure 4.10

Effect of an Increase in Dividend Income or a Decrease in Taxes

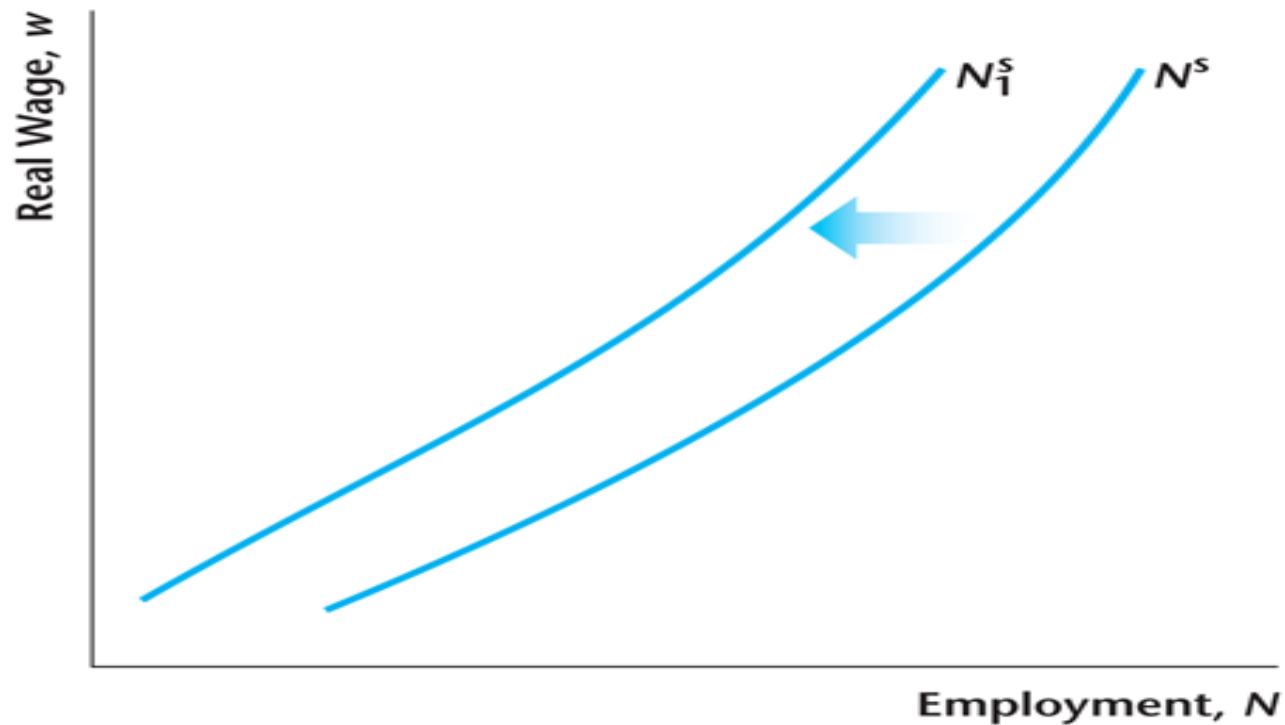
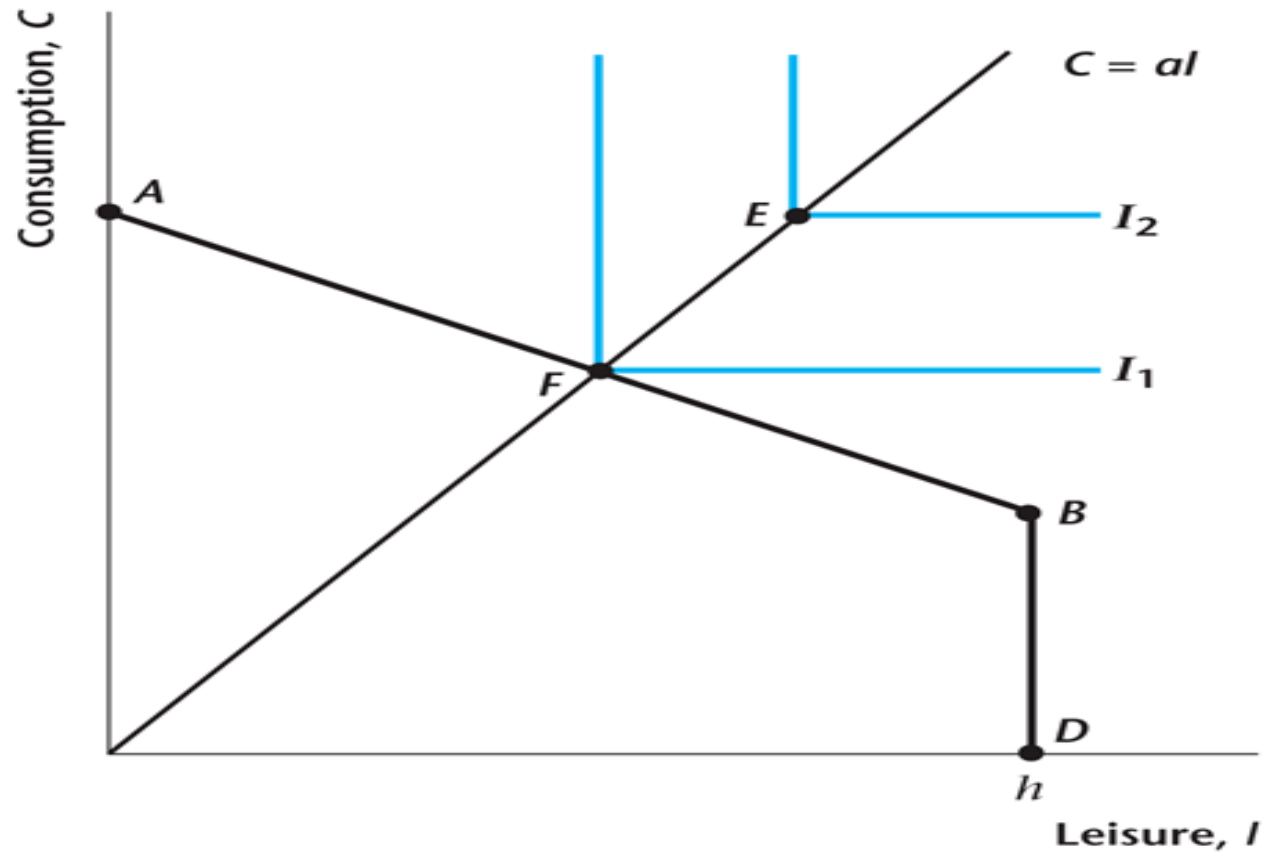


Figure 4.11
Perfect Complements



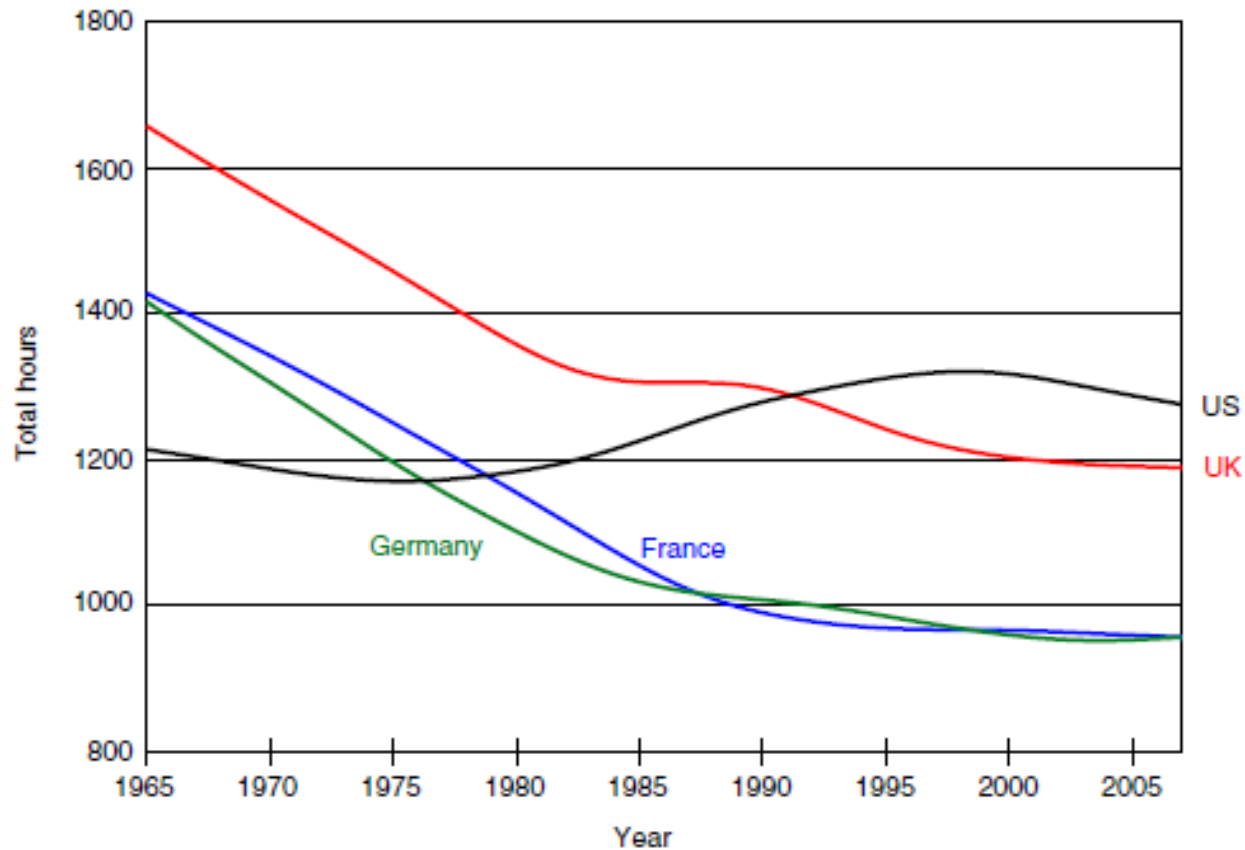
Practice question

Suppose that Chris works part time as a cashier and can earn a higher wage rate for working overtime. He receives a real wage of w_1 for the first q hours of his normal work, and for hours worked more than q , he receives w_2 , where $w_2 > w_1$.

Assume that there is no tax or non-labor income, and he is free to choose hours of work. His preference over c and l are given by $u(c, l) = \min(c, 2l)$.

Illustrate your answers on a diagram. Draw the student's budget constraint and show his or her possible optimal choice of consumption and leisure.

Case study: Why Americans work more than Europeans?



Case study: Why Americans work more than Europeans?

- Today, hours worked in Europe are almost 30 percent lower than in the U.S.
- Cross-country variation in tax systems is one of the most popular proposed explanations for the observed discrepancy in work hours between the US and Europe. [Prescott (2004)]
- Holter, Chakraborty, and Stepanchuk (2015) believe that divorce rates also play a role, particularly for women's labor supply.